

Computability: Exercise 2

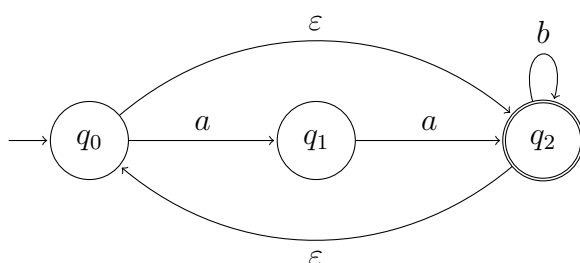
Due: April 19, 2026

Question 1

Given a DFA $\mathcal{A} = \langle Q, \Sigma, \delta, q_0, F \rangle$, define an NFA $\mathcal{B}$ for which $L(\mathcal{B}) = L(\mathcal{A})$. Deduce that $\text{REG} \subseteq \text{NREG}$.

Question 2

- a. Draw the following NFA without ϵ - transitions:



- b. Let $L \in \text{REG}$ be a language over an alphabet Σ , and show that the Kleene closure L^* in NREG .
- c. Prove part (b) without ϵ - transitions.

Question 3

Set $\Sigma = \{0, 1, \dots, 9\}$ and for any $k \in \mathbb{N}$ define

$$L_k := \{w \in \Sigma^{2k} \mid \exists i \in \{1, \dots, k\} \text{ such that } w_i = w_{k+i}\},$$

that is, the language of all words from length $2k$ that their first half is equal to their second half at some position i (i.e., there exists $i \in [k]$ ¹ such that $w_i = w_{k+i}$). In addition, for every $k \in \mathbb{N}$ and $i \in [k]$, we define

$$L_k^i := \{w \in \Sigma^{2k} : w_i = w_{k+i}\},$$

that is, the language of all words of length $2k$ that their first half is equal to their second half at position i .

- a. Show that for every $k \in \mathbb{N}$ and $i \in [k]$, L_k^i is regular.
- b. Show that there exists for each k an NFA that recognizes L_k , which has a number of states of order $O(k^2)$.
- c. Show that any k and any DFA $\mathcal{A}$ that decides L_k , $\mathcal{A}$ has at least 2^k states.

¹We denote $[k] = \{1, \dots, k\}$.

Question 4

Let L be a regular language over an alphabet Σ , and show that

$$L^R := \{w^R : w \in L\} \in \text{NREG},$$

where for a word $w = w_1w_2 \cdots w_k$ we define $w^R := w_kw_{k-1} \cdots w_1$.

Question 5

Let $\mathcal{A} = \langle Q, \Sigma, \delta, q_0, F \rangle$ be a DFA.

- a. Prove that if $\mathcal{A}$ accepts a word w for which $|w| \geq |Q|$ then there exist a pair of indices $0 \leq i < j \leq |w|$ such that $\delta^*(q_0, w_{\leq i}) = \delta^*(q_0, w_{\leq j})$, where $w_{\leq i}$ is the prefix of w that consists of its first i letters.
- b. Show that if $L(\mathcal{A})$ contains a word w for which $|w| \geq |Q|$ then it is actually infinite.
Hint: by the previous section $\mathcal{A}$ visits some state q at least twice when it runs on w . Find a way to add letters to w so that $\mathcal{A}$ visits q even more times on the larger word, while still reaching an accepting state at the end.